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# Factors affecting the distribution of benthic macrofauna in estuaries contaminated by urban runoff

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**Abstract**

Contaminants derived from urban runoff have been shown to accumulate in estuarine sediments, reaching concentrations potentially capable of causing biological effects. Demonstration of effects, however, is difficult due to strong natural environmental gradients and the effects of past or present point-sources of contamination. We used multivariate methods to test two hypotheses relating to the effects of urban-derived contaminants on estuarine benthic communities. First, that patterns of distribution and abundance of benthic invertebrates in two urbanised estuaries would be different from those in two non-urbanised estuaries. Second, that the distributions of benthic invertebrates within and among the four estuaries would be related to those of urban-derived contaminants. Concentrations of contaminants were larger in estuaries with urbanised catchments and concentrations of Cu, Pb, Zn and DDT in some samples exceeded those at which biological effects may be expected to appear. Tests of differences in composition of benthic communities among estuaries showed that the two urban estuaries were not significantly different, but that they differed from both rural estuaries, which also differed from each other. Distributions of benthic invertebrates were significantly related to those of environmental variables, and were ordinated along axes that correlated with both natural environmental variables (nature of the sediment, position in estuary) and contaminants. Differences in faunas between the urban and non-urban estuaries were not, however, clear-cut and nor were relationships between faunal assemblages and environmental

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variables (including contaminants) consistent between two times of sampling.
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*Keywords:* Estuary; Stormwater; Sediment; Contaminant; Macrofauna; New Zealand

## 1. Introduction

Stormwater runoff from urban areas is a significant source of organic and inorganic contaminants to receiving environments (Förstner & Wittmann, 1979). These contaminants derive from vehicles and other sources, and include heavy metals, notably copper, lead and zinc, and organic compounds, notably polycyclic aromatic hydrocarbons (PAH) (Velinsky, Wade, Schlekat, McGee, & Presley, 1994; Williamson, 1991). Since much of the contaminant load is associated with suspended particulate material (Förstner & Wittmann, 1979), contaminants will tend to accumulate in environments where particulate material is deposited and remains for relatively long periods. The eventual, long-term sinks for this material are sheltered coastal environments, particularly estuaries. Contaminants derived from stormwater may reach concentrations in estuarine sediments potentially capable of causing biological effects (e.g. as identified by the sediment quality guidelines of Long, MacDonald, Smith, & Calder, 1995). Such levels have been identified, for example, in urbanised estuaries in New Zealand (Deely & Ferguson, 1994; Morrisey, Williamson, & Roper, 1997; Morrisey, Williamson, Van Dam, & Lee, 2000).

Input and accumulation of contaminants from runoff in receiving environments can be reduced by treatment technology, such as detention ponds, but their effectiveness with respect to protection of downstream ecosystems has not been demonstrated (Maxted & Shaver, 1997). Consequently, demonstration of ecological effects of contaminants derived from runoff is necessary in assessing the scale of the problem and as a precursor to assessing the effectiveness of treatment technology in mitigating these effects. Although biological effects associated with contaminants derived from stormwater have been demonstrated in freshwater streams (e.g. Pedersen & Perkins, 1986), this task has proved more difficult in estuarine and coastal environments (Pearce, 1995; Maxon, Barnett, & Diener, 1997; Morrisey et al.; Nipper et al., 1998). This is because of the relatively weak effects produced by contamination derived from stormwater compared to those produced by natural estuarine gradients (such as salinity and sediment texture: Diaz, 1989), physical disturbances and, in many cases, industrial contamination.

We emphasise that while concentrations of metals and other contaminants in these estuaries may exceed criteria at which biological effects begin to manifest (Morrisey et al., 2000), such as the effects range-low (ER-L) values of Long et al. (1995), they do not reach concentrations at which effects would be expected, such as the effects range-median (ER-M: Long et al., 1995). What is needed are tools that will allow more subtle effects to be identified while concentrations of contaminants are still relatively low, before they reach levels at which severe effects appear and expensive, retrospective action is required to mitigate them. Multivariate statistical methods

provide a potential tool for unravelling the effects of multiple environmental variables on the distributions of organisms (Clarke, 1999; Sparks, Scott, & Clarke, 1999).

We present the results of a study of urban and rural estuaries in New Zealand, in which we tested two hypotheses concerning relationships between the distributions of environmental variables, including contaminants derived from runoff, and those of communities of sediment-living invertebrates. First, that patterns of distribution and abundance of communities of benthic invertebrates in two urbanised estuaries would be different from those in two non-urbanised estuaries. This was tested using a multivariate test of the significance of differences among groups of faunal samples derived from the respective estuaries. Second, that the distribution of benthic invertebrates within and among the four estuaries would be related to that of stormwater-derived contaminants. This was tested using multivariate correlations of benthic communities with a suite of environmental variables (including contaminants) presumed, a priori, to influence their distribution.

## 2. Methods

### 2.1. Sampling

Four estuaries (or subestuaries of larger estuarine systems) were selected in the vicinity of Auckland, North Island, New Zealand (Fig. 1). The catchments of two of these, Te Matuku Creek on Waiheke Island and Paremoremo Creek in the upper Waitemata Harbour, are in predominantly agricultural (grazing) use or native bush (Table 1). The catchments of the other two, Hellyers Creek in the upper Waitemata Harbour and Pakuranga Creek in the Tamaki Estuary, are in predominantly residential use (Table 1). These estuaries are hereafter referred to as 'Waiheke', 'Paremoremo', 'Hellyers' and 'Pakuranga', respectively. All four estuaries are narrow, steep-sided, drowned river valleys having free exchange with the sea or larger estuarine system. No form of runoff treatment is in place in any of these catchments. It is also important to note that there are no combined stormwater–sewer outfalls in any of the estuaries, nor were there any identifiable point sources of contaminants that would be expected to have strong, local effects on biota. The two urbanised estuaries were selected on the basis of the following criteria: no history of industrial discharges to natural waterways; no combined sewer overflows; minimal boating activity (few moorings, no marinas, no major boat-launching facilities); a sufficiently long history of urban development (pre-1970) that there is likely to be a significant change in concentrations of metals in sediments over that expected from background and outside sources. More than two estuaries within each category (urban, non-urban) would have increased the power of statistical tests among categories and strengthened inferences about the ecological effects of urbanisation on estuaries. It was not, however, possible to find additional, suitable estuaries because of major differences in geomorphology or the presence of strong current or historical point-sources of contamination.

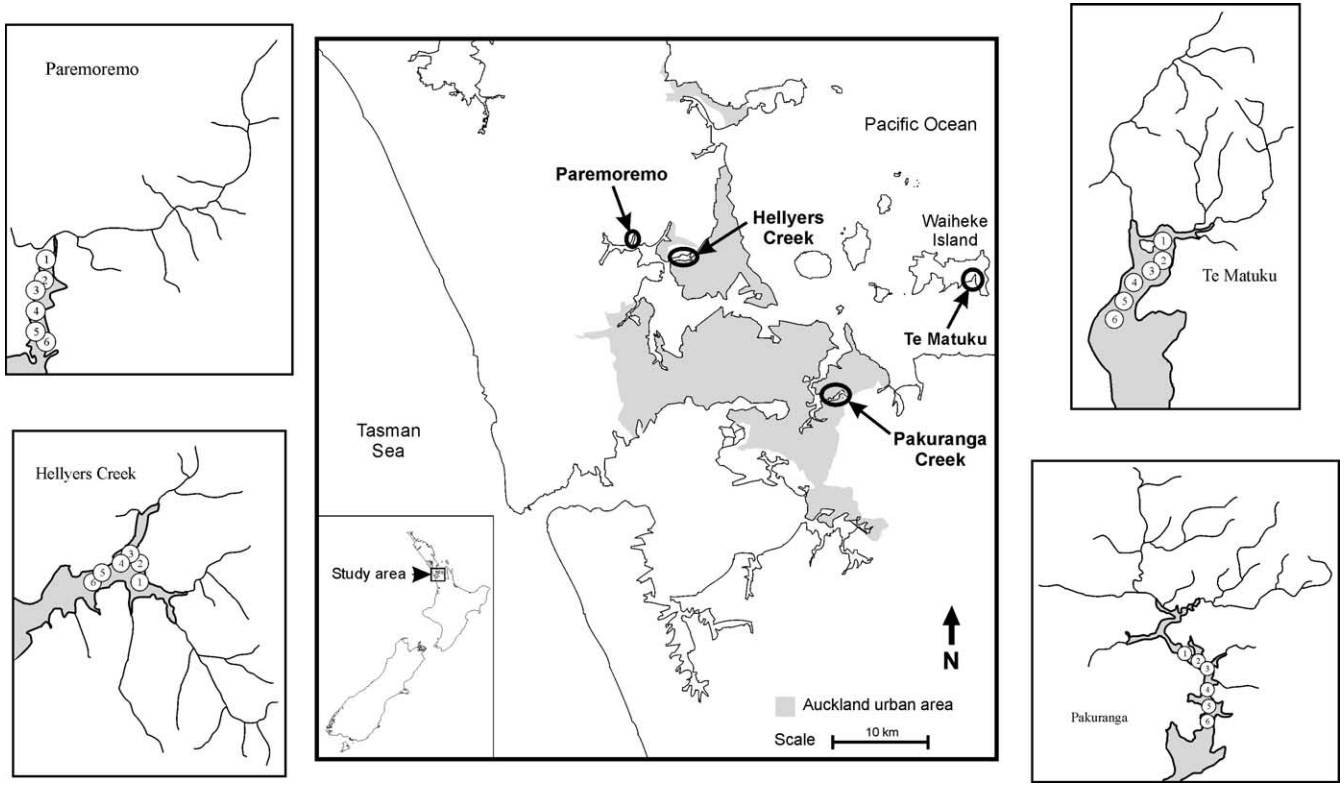


Fig. 1. Map showing the location of the estuaries studied and of the sampling sites within each estuary.

Table 1
Location, size and land-use characteristics of the catchments of the four estuaries

| | Estuary | | | |
|---|---|---|---|---|
| | Waiheke | Paremoremo | Hellyers | Pakuranga |
| Location | 36° 50′S 175° 8′E | 36° 46′S 174° 38′E | 36° 47′S 174° 42′E | 36° 56′S 174° 53′E |
| Area of estuary (ha) | 93 | 42 | 113 | 95 |
| Area of catchment (ha) | 1253 | 1221 | 941 | 2545 |
| *Land use (%)* | | | | |
| Residential | 0 | 0 | 77 | 69 |
| Light industrial | 0 | 0 | 1 | 12 |
| Commercial | 0 | 0 | Small, local centres only | 1 |
| Pasture | 55 | 61 | 0 | 18 |
| Bush | 45 | 39 | 22 | 0 |

Inputs of heavy metals to estuaries with non-urban catchments were presumed to be at background (natural) levels. There is no information available on inputs of organic contaminants to non-urban estuaries in New Zealand but, given the agricultural use of parts of the catchments of Paremoremo and Waiheke, contamination with organochlorine pesticides might be expected. Studies of concentrations of these compounds in other non-urban estuaries in New Zealand, in contrast, indicate levels below detection limits (0.01–0.20 ng $g^{-1}$ dry weight; (Ministry for the Environment, 1998) and well below the guidelines of Long et al. (1995).

Six sites, measuring 50 m × 20 m (parallel and perpendicular to the orientation of the low-tide channel, respectively), were selected along the lengths of each of the estuaries. Sites were located just above the bank of the low-tide channel. The relative positions of the sites are shown in Fig. 1, with Site 1 being the furthest upstream and Site 6 the furthest downstream in each estuary. The position of each site within an estuary was expressed as the distance of the site from the mouth of the estuary as a percentage of the total length of the estuary. This was used as a surrogate for environmental factors that vary along the estuarine gradient but which are difficult to summarise in a single variable. For example, salinity of the overlying water varies with position in the estuary, but also with the state of the tide and volume of freshwater input, making the use of a single measure of salinity inappropriate. The sediment in all of the sites consisted of soft mud (i.e. silt plus clay) or fine-sandy mud (range 25–89% mud). The distribution of sites was not intended to provide representation of the whole of each estuary but to ensure some correspondence in the positions of sites along environmental gradients (salinity, type of sediment, concentrations of contaminants, etc.) among the four estuaries.

Four sampling points (hereafter referred to as "plots") were identified within each site using random Cartesian coordinates with the constraint that plots could not be closer than 15 m, to ensure adequate dispersal of sampling within the site and to prevent trampling of sampling areas. Four core samples (13 cm diameter, 15 cm deep) were collected at each of the four plots within each site. Two of these cores were spaced about 0.5 m apart, one to be used for measurement of chemical and physical variables (see below) and the other for enumeration of invertebrates. Two

further cores were taken 2 m apart, either side of the first two cores, also for enumeration of invertebrates. Thus, three cores were collected in each plot to provide an estimate of small-scale (1–2 m) variation in faunal abundance and diversity. Resources did not permit triplicate sampling of physicochemical variables.

Sampling of sites in Hellyers and Pakuranga was done in May 1995 and in Paremoremo and Waiheke in March 1996. Because we were seeking general relationships between the composition of the benthic faunas of the estuaries and the environmental variables, this difference in time of sampling was not considered to be of overriding significance. To test this assumption of the generality over time of any relationships identified, all four estuaries were resampled in February 1997. Because of limited resources, only two sites in each estuary were resampled and organic contaminants (see below) were not measured.

Cores for enumeration of invertebrates were sieved immediately through 500-μm mesh, preserved in 70% isopropyl alcohol with Rose Bengal stain and later identified to the lowest practical taxonomic level. Chemistry cores were kept refrigerated until processing (within 24 h). Each core was placed in a plastic tray and homogenised with a stainless steel spatula. Subsamples were taken for analysis of trace metals and organic contaminants (see below). Further subsamples were taken for analyses of particle-size distribution (% mud, sand, gravel), sediment consistency (see below), % volatile organic matter (VOM) and for extraction of porewater.

### 2.2. Chemical and physical analyses

Determinands were selected on the basis of their common presence in stormwater runoff and their occurrence in previously-studied, muddy estuaries in New Zealand. For analyses of trace metals (Cu, Fe, Pb, Zn), approximately 1 g of wet sediment was placed in a stoppered, 50-ml polypropylene centrifuge-tube. The sediment in the tubes was dried at 105 °C overnight and weighed. Forty millilitres of 2-M HCl were added to each tube to extract the reactive fraction of metals. This fraction is associated with phases actively involved in early diagenesis, i.e. adsorbed or coprecipitated with FeS, $FeCO_3$, $CaCO_3$, FeOOH, MnOOH and degradable organic matter (Williamson, Mol-Krijnen, & Van Dam, 1995). It will include the bioavailable fraction, either that which is available at the time of sampling or after diagenetic changes in sediment characteristics, and probably much of the anthropogenic input. The tubes were then placed on their sides on a shaking table at 100 rpm for 24 h. Samples were centrifuged at $1500 \times g$ for 15 min and the supernatant analysed for Cu, Fe, Pb and Zn by atomic absorption spectrometry (Perkin Elmer 3110). Concentrations of metals were expressed as mg $g^{-1}$.

Organic contaminants analysed were 15 three-to-six ring polycyclic aromatic hydrocarbons (PAH), organochlorine pesticides (*o*,*p*′- and *p*,*p*′- isomers of DDT, DDD, DDE, aldrin, lindane, dieldrin, and six chlordane compounds), and 28 dichloro- to decachloro- polychlorinated biphenyl congeners (PCB). Sediments were air-dried, sieved (500 μm), spiked with analytical surrogates, then extracted with dichloromethane by Accelerated Solvent Extraction. Extracts were cleaned and fractionated using silica gel chromatography before analysis by capillary GC-MS (PAH)

and dual column capillary GC-ECD (PCBs and pesticides). Detection limits were 0.05–0.1 ng g-1 for pesticides and PCB and 0.1–0.5 ng $g^{-1}$ for PAH. Quality assurance included monitoring of surrogate recoveries, analysis of blanks and sample duplicates (ca. 10% of samples analysed), and analysis of standard reference material SRM1941a (US National Institute of Standards and Technology). Concentrations were expressed as ng $g^{-1}$ dry weight (105° C) of sieved sediment.

Porewater was extracted by centrifugation ($9000 \times g$ for 30 min) from two of the chemistry cores from each site. For measurement of concentrations of metals (Cu, Pb and Zn), the porewater was acidified with $HNO_3$ (Aristar quality) to 0.5% and analysed by AAS (Perkin Elmer Zeeman 4100ZL). Concentrations of metals were at or below detection limits (0.02, 0.05, and 0.001 μg $l^{-1}$, for Cu, Pb and Zn, respectively) and were, therefore, not used in subsequent multivariate analyses.

Sediment grain size (% gravel, sand, mud) was estimated by sieving (Buchanan, 1984). The consistency of the sediment in each sample was assigned a value on an arbitrary scale of 1–4, where 1 and 4 represented fluid and stiff mud, respectively. Dry weight (105 °C) of the sediment as a percentage of the freshly collected wet weight was also used as a descriptive variable. Percent VOM was measured after combustion at 400 °C for 18 h.

### 2.3. Statistical analyses

Total numbers of animals and numbers of invertebrate taxa were compared among estuaries and among sites and plots within each of the four estuaries by 3-factor ANOVA (separate analyses for the two times of sampling), with 'site' and 'plot' as a random factors nested in the fixed factor 'estuary'. Differences among estuaries were identified using Scheffe's test. Where heterogeneity of variances was found (Cochran's test), data were $\log_e(x+1)$-transformed.

Differences in invertebrate community composition at each site in each of the estuaries and for each sampling period were examined using non-metric multidimensional scaling (NMDS) ordinations (using the software package PRIMER; Clarke, 1993). Ordinations were based on the Bray–Curtis similarity index using untransformed, 4th-root ($\sqrt{\sqrt{}}$) transformed and presence/absence community data for each site within each estuary. Transformations play an important role in ordination procedures by defining the balance between contributions from common and rarer taxa in the measure of similarity of two samples (Clarke & Warwick, 1994). In the present case, the NMDS ordinations of untransformed, $\sqrt{\sqrt{}}$-transformed and presence/absence community data were all similar, indicating that the information provided by the untransformed data was not being affected by a few dominant taxa. Untransformed data were therefore used for all subsequent analyses.

Significance testing for differences in community composition among different estuaries and among sites within each estuary was done using the ANOSIM ("analysis of similarities") randomisation test, based on rank similarities of the samples (Clarke, 1993). Two-factor, nested comparisons among the four estuaries and among sites within estuaries were made using plots as replicates and pooling cores within each plot. In a second set of single-factor analyses, sites within each estuary

were compared using plots as replicates, pooling cores and analysing each estuary separately. Two-way, crossed ANOSIM tests were also used to examine differences in community composition between the two sampling periods and among sites within each estuary. The SIMPER ("similarity percentages"; Clarke, 1993) routine was used to identify the taxa that made the greatest contribution to any differences among estuaries and among sites within estuaries identified in the NMDS plots and ANOSIM tests (Clarke & Warwick, 1994).

Canonical correspondence analysis (CCA, using the software package CANOCO; ter Braak, 1986, 1988, 1990) was used to estimate the amount of variation in the faunal data explained by the set of measured environmental variables: sediment characteristics (grain size, % VOM, sediment consistency and % dry weight), concentrations of metals (copper, zinc, lead and iron) and organic contaminants (PAH, dieldrin, chlordane, DDT and PCB, all normalised to concentrations of organic matter), and the percentage distance of each site along each of the respective estuaries. In canonical correspondence analysis, the axes within which the biological data are ordinated are constrained to be linear combinations of the environmental variables. The biological data are plotted within these axes, both as individual taxa and as assemblages of taxa present in individual samples. The relative positions of the points representing taxa and samples indicate the major patterns in the composition of communities to the extent that these can be 'explained' by the measured environmental variables. Environmental variables are also plotted, and the positions of taxa in the plot relative to those of environmental variables indicate the positions of the various taxa along each individual environmental gradient.

Neither the environmental nor the faunal data were transformed (environmental data were standardised). The effect of "geographical location" on benthic community composition was removed by specifying the map grid reference (New Zealand Map Grid) of each site as a covariable in the analysis (Borcard, Legendre, & Drapeau, 1992). Environmental variables that were highly correlated with other environmental variables (i.e. variance inflation factor $> 20$: ter Braak, 1988; ter Braak & Wiertz, 1994) were removed from the analyses one at a time, in order of lowest correlations with either of the correspondence axes 1 and 2. The significance of relationships between the constraining environmental variables and the patterns of differences in benthic community composition extracted by CCA was tested using CANOCO's Monte Carlo permutation test (ter Braak, 1987). Significance probabilities are given for the overall "trace" (sum) of the eigenvalues for the constrained (canonical) axes.

## 3. Results

### 3.1. Distribution of physical and chemical variables

As expected, the distributions of several of the environmental variables measured in the present study were characterised by gradients along each of the estuaries, as indicated by their correlations with % distance (Tables 2 and 3). Concentrations of

Table 2
Mean values (±SE, $n=4$) for selected environmental variables at the sites in each of the four estuaries

| Estuary | Site | % mud | % sand | % VOM | Cu (μg g$^{-1}$) | Pb (μg g$^{-1}$) | Zn (μg g$^{-1}$) | PAH (ng g$^{-1}$) | PCB (ng g$^{-1}$) | DDT (ng g$^{-1}$) |
|---|---|---|---|---|---|---|---|---|---|---|
| (a) *1995/1996* | | | | | | | | | | |
| Hel | 1 | 82.2 (2.4) | 17.8 (2.4) | 4.8 (0.4) | 25.6 (1.7) | 76.3 (6.4) | 98.6 (4.7) | 787.6 (29.7) | 4.68 (0.40) | 2.84 (0.14) |
| | 2 | 81.9 (7.4) | 18.1 (7.4) | 4.7 (0.5) | 23.2 (1.9) | 57.4 (4.9) | 93.4 (7.1) | 850.4 (48.0) | 4.54 (0.35) | 3.43 (0.34) |
| | 3 | 80.8 (4.2) | 19.2 (4.2) | 4.8 (0.3) | 23.6 (3.9) | 54.8 (5.8) | 98.7 (7.0) | 892.7 (42.3) | 5.59 (0.71) | 3.12 (0.29) |
| | 4 | 60.7 (5.5) | 39.1 (5.4) | 3.9 (0.3) | 17.9 (1.5) | 31.3 (2.4) | 83.8 (2.8) | 701.2 (48.7) | 3.77 (0.34) | 2.40 (0.09) |
| | 5 | 70.3 (8.3) | 29.7 (8.3) | 4.1 (0.4) | 16.7 (1.3) | 32.0 (3.7) | 83.5 (6.3) | 704.2 (58.6) | 4.17 (0.50) | 3.68 (0.92) |
| | 6 | 65.7 (5.2) | 34.3 (5.1) | 4.5 (0.5) | 17.1 (1.4) | 31.1 (2.9) | 85.4 (3.1) | 741.5 (29.5) | 4.80 (0.27) | 2.96 (0.25) |
| Pak | 1 | 78.6 (8.3) | 21.3 (8.2) | 4.9 (0.6) | 32.4 (3.4) | 63.0 (5.7) | 215.2 (8.7) | 891.7 (24.5) | 6.69 (0.17) | 2.36 (0.07) |
| | 2 | 69.0 (9.0) | 28.9 (7.2) | 4.7 (0.5) | 25.0 (3.2) | 43.4 (8.3) | 165.8 (21.2) | 653.6 (37.7) | 5.55 (0.49) | 1.83 (0.25) |
| | 3 | 88.7 (1.8) | 11.3 (1.8) | 5.4 (0.4) | 32.7 (2.5) | 56.3 (2.5) | 220.1 (12.7) | 861.0 (23.4) | 7.70 (0.36) | 3.34 (0.33) |
| | 4 | 64.2 (9.9) | 35.7 (9.9) | 4.0 (0.4) | 19.2 (1.9) | 27.9 (2.0) | 141.2 (9.8) | 581.3 (46.9) | 4.17 (0.42) | 1.88 (0.33) |
| | 5 | 32.5 (2.1) | 62.5 (3.2) | 2.8 (0.2) | 14.9 (0.6) | 21.7 (2.2) | 127.0 (4.5) | 406.4 (16.8) | 2.86 (0.13) | 1.27 (0.14) |
| | 6 | 25.2 (1.5) | 72.8 (0.7) | 2.3 (0.1) | 12.4 (0.6) | 19.9 (0.8) | 120.5 (3.3) | 351.3 (8.3) | 1.94 (0.16) | 1.02 (0.03) |
| Par | 1 | 83.9 (4.1) | 16.1 (4.1) | 5.5 (0.3) | 13.3 (0.3) | 11.0 (1.4) | 60.7 (2.0) | 509.0 (43.6) | 1.47 (0.09) | 2.84 (0.23) |
| | 2 | 73.5 (10.9) | 25.4 (10.6) | 4.8 (0.7) | 11.9 (1.5) | 7.9 (3.2) | 48.7 (2.8) | 438.2 (55.7) | 1.48 (0.23) | 2.27 (0.27) |
| | 3 | 95.6 (1.6) | 4.4 (1.6) | 5.5 (0.1) | 20.0 ( 1.2) | 18.2 (3.4) | 73.7 (2.7) | 617.5 (27.2) | 2.44 (0.21) | 3.32 (0.30) |
| | 4 | 36.8 (3.3) | 63.0 (3.2) | 2.3 (0.2) | 9.0 (0.3) | 9.1 (1.7) | 38.4 (0.9) | 191.6 (18.3) | 0.76 (0.08) | 1.39 (0.10) |
| | 5 | 38.1 (3.9) | 61.9 (3.9) | 2.7 (0.2) | 10.1 (0.2) | 14.8 (1.3) | 65.7 (2.0) | 222.1 (30.2) | 0.89 (0.14) | 1.59 (0.21) |
| | 6 | 52.2 (10.7) | 46.5 (9.9) | 4.1 (0.8) | 10.9( 1.5) | 8.1 (2.7) | 51.6 (0.6) | 470.5 (78.4) | 1.53 (0.31) | 2.57 (0.29) |
| Wai | 1 | 58.4 (8.7) | 41.6 (8.7) | 5.3 (1.0) | 6.0 (0.4) | 7.3 (1.4) | 27.5 (2.5) | 121.4 (16.3) | 0.06 (0.02) | 0.50 (0.32) |
| | 2 | 46.8 (4.2) | 44.2 (3.5) | 3.8 (0.2) | 5.2 (0.5) | 2.6 (1.5) | 27.6 (0.8) | 70.2 (12.4) | 0.04 (0.02) | 0.15 (0.01) |
| | 3 | 33.5 (8.0) | 49.2 (5.6) | 2.9 (0.4) | 4.4 (0.5) | 0.0 (0.0) | 22.2 (1.2) | 83.3 (16.8) | 0.03 (0.03) | 0.08 (0.04) |
| | 4 | 45.6 (6.2) | 52.8 (5.2) | 3.1 (0.2) | 5.7 (0.3) | 0.0 (0.0) | 30.7 (0.5) | 95.2 (14.6) | 0.06 (0.04) | 0.12 (0.03) |
| | 5 | 78.3 (6.0) | 18.2 (4.5) | 5.7 (0.7) | 7.8 (0.5) | 1.5 (1.5) | 41.4 (1.8) | 125.7 (11.6) | 0.06 (0.02) | 0.40 (0.08) |
| | 6 | 26.6 (2.0) | 51.5 (2.7) | 2.5 (0.1) | 5.8 (0.6) | 0.0 (0.0) | 21.0 (0.8) | 57.9 (5.9) | 0.00 (0.00) | 0.13 (0.00) |
| (b) *1997* | | | | | | | | | | |
| Hel | 3 | 79.8 (3.6) | 20.2 (3.6) | 8.0 (0.9) | 14.3(1.0) | 72.9 (2.7) | 111.0 (6.0) | nr | nr | nr |
| | 5 | 47.8 (7.2) | 52.2 (7.1) | 4.5 (0.6) | 8.4 (1.3) | 51.1 (1.9) | 78.5 (6.5) | nr | nr | nr |
| Pak | 2 | 51.3 (5.9) | 43.1 (4.2) | 4.6 (0.5) | 10.8 (2.4) | 53.3 (8.3) | 134.9 (25.1) | nr | nr | nr |
| | 5 | 29.7 (1.0) | 69.0 (1.2) | 3.6 (0.1) | 9.8 (0.5) | 47.3 (2.1) | 138.0 (7.3) | nr | nr | nr |
| Par | 3 | 97.9 (0.3) | 2.1 (0.3) | 9.7 (0.4) | 14.1( 0.6) | 58.9 (1.2) | 71.8 (0.5) | nr | nr | nr |
| | 6 | 38.7 (9.8) | 58.8 (8.8) | 5.1 (1.0) | 5.9 (1.4) | 40.8 (0.8) | 49.3 (3.7) | nr | nr | nr |
| Wai | 4 | 39.7 (2.7) | 58.6 (2.2) | 3.2 (0.1) | 1.0 (0.1) | 25.8 (1.5) | 26.9 (1.2) | nr | nr | nr |
| | 6 | 47.3 (4.8) | 27.5 (2.1) | 3.4 (0.3) | 0.4 (0.2) | 31.1 (2.2) | 30.3 (1.4) | nr | nr | nr |

'% mud', '% sand' and '% VOM' are percentages of mud, sand and volatile organic matter, respectively, in the sediment; concentrations of contaminants (Cu, Pb, Zn, PAH and DDT) are expressed per weight of dry sediment and are not normalised to organic matter content. 'nr', not recorded. See text for details of sampling. Abbreviations of estuary names: Hel = Hellyers, Pak = Pakuranga, Par = Paremoremo and Wai = Waiheke.

heavy metals decreased with distance downstream in all estuaries. Concentrations of organic contaminants showed this pattern of distribution in Pakuranga but not in the other estuaries. Concentrations of organic contaminants correlated with % VOM in all estuaries (Table 3).

### 3.2. Distributions of benthic invertebrates

There were no significant differences in numbers of taxa among estuaries in 1995/1996 (Table 4). Mean numbers of taxa per core (Table 5) were largest in Waiheke and Paremoremo and ranged from 3.0 to 12.8, 5.9 to 9.8, 3.9 to 8.5 and 3.5 to 7.7 for

Table 3
Pearson correlation coefficients for comparisons between distance and other environmental variables and between% VOM and organic contaminants (across all cores and sites, $n = 24$) in 1995/1996

| | Hellyers | | Pakuranga | | Paremoremo | | Waiheke | |
|---|---|---|---|---|---|---|---|---|
| | *r* | *P* | *r* | *P* | *r* | *P* | *r* | *P* |
| *% distance vs* | | | | | | | | |
| % sand | −0.499 | <0.05 | −0.771 | <0.001 | −0.620 | <0.01 | 0.084 | ns |
| % mud | 0.498 | <0.05 | 0.761 | <0.001 | 0.618 | <0.01 | 0.105 | ns |
| % VOM | 0.268 | ns | 0.746 | <0.001 | 0.536 | <0.01 | 0.235 | ns |
| copper | 0.611 | <0.01 | 0.762 | <0.001 | 0.347 | ns | −0.333 | ns |
| zinc | 0.425 | <0.05 | 0.690 | <0.001 | 0.095 | ns | −0.140 | ns |
| lead | 0.845 | <0.001 | 0.778 | <0.001 | 0.014 | ns | 0.593 | <0.01 |
| iron | 0.505 | <0.05 | 0.213 | ns | −0.102 | ns | −0.134 | ns |
| PAH | 0.315 | ns | 0.836 | <0.001 | 0.351 | ns | 0.113 | ns |
| dieldrin | 0.301 | ns | 0.460 | <0.05 | −0.050 | ns | – | |
| chlordane | 0.486 | <0.05 | 0.763 | <0.001 | 0.230 | ns | −0.205 | ns |
| DDT | −0.003 | ns | 0.553 | <0.01 | 0.315 | ns | 0.139 | ns |
| PCB | 0.063 | ns | 0.802 | <0.001 | 0.245 | ns | 0.147 | ns |
| *% VOM vs:* | | | | | | | | |
| PAH | 0.73 | <0.001 | 0.964 | <0.001 | 0.958 | <0.001 | 0.476 | <0.05 |
| dieldrin | 0.687 | <0.001 | 0.755 | <0.001 | 0.832 | <0.001 | – | |
| chlordane | 0.773 | <0.001 | 0.963 | <0.001 | 0.895 | <0.001 | 0.405 | ns |
| DDT | 0.864 | <0.001 | 0.842 | <0.001 | 0.884 | <0.001 | 0.76 | <0.001 |
| PCB | 0.572 | <0.01 | 0.968 | <0.001 | 0.857 | <0.001 | 0.718 | <0.001 |

Table 4
Summary of results of analyses of variance for number of taxa and number of individuals among estuaries, sites and plots in 1995/1996

| Factor | df | No. taxa | | No. individuals | |
|---|---|---|---|---|---|
| | | *F* | *P* | *F* | *P* |
| Estuary | 3 | 1.38 | 0.278 | 6.27 | 0.004 |
| Site(Estuary) | 20 | 9.22 | <0.0001 | 5.76 | <0.0001 |
| Plot(Site(Estuary)) | 72 | 2.13 | <0.0001 | 1.74 | 0.002 |
| Residual | 192 | | | | |
| Scheffe's test on factor 'Estuary' | | | | Wai Pak Hel Par | |

Numbers of individuals were $\log_e(x+1)$−transformed to remove heterogeneity of variances. Differences among estuaries, as identified by Scheffe's multiple comparison test, are shown for the numbers of individuals (abbreviations of estuary names as in Table 2). Equivalent tests among sites within estuaries and

Waiheke, Paremoremo, Pakuranga and Hellyers, respectively. Mean numbers of individuals per core differed significantly among estuaries (Table 4) and were largest in Paremoremo, much smaller in Waiheke and Hellyers, and smallest in Pakuranga (Table 5: ranges 46.0–240.7, 6.1–56.3, 9.8–45.8 and 8.3–44.5 for Paremoremo, Waiheke, Hellyers and Pakuranga, respectively). The multiple comparison tests for the

Table 5
Mean (SE, $n=12$) numbers of taxa, total individuals and individuals of selected taxa at each site in each estuary (abbreviations of estuary names as in Table 2)

| Estuary | Site | Taxa | Individuals | Capitellids | Corophiid sp. 1 | *Helice crassa* | Nereids | Oligochaetes | Paraonid sp. 1 | Polydorids |
|---|---|---|---|---|---|---|---|---|---|---|
| *1995/1996* | | | | | | | | | | |
| Hel | 1 | 3.5 (0.60) | 9.8 (1.45) | 0.2 (0.11) | 1.5 (0.70) | 2.8 (0.91) | 1.0 (0.51) | 0.1 (0.08) | 0.1 (0.08) | 1.9 (0.67) |
| | 2 | 4.9 (0.71) | 38.7 (10.10) | 0.3 (0.14) | 25.7 (9.86) | 5.3 (1.10) | 0.8 (0.35) | 1.1 (0.66) | 0.6 (0.23) | 3.8 (1.30) |
| | 3 | 5.6 (0.53) | 38.4 (13.29) | 0.9 (0.34) | 20.5 (12.89) | 0.3 (0.18) | 3.0 (0.85) | 1.1 (0.60) | 0.3 (0.19) | 9.5 (1.99) |
| | 4 | 6.3 (0.48) | 34.3 (7.02) | 0.7 (0.43) | 5.8 (3.06) | 0.4 (0.29) | 3.6 (1.23) | 1.9 (1.47) | 2.6 (0.67) | 14.7 (4.32) |
| | 5 | 6.2 (0.47) | 30.7 (4.91) | 2.8 (0.71) | 3.3 (1.88) | 0.4 (0.26) | 1.4 (0.29) | 0.3 (0.13) | 3.1 (0.67) | 12.9 (3.79) |
| | 6 | 7.7 (0.54) | 45.8 (3.98) | 7.7 (1.83) | 0.3 (0.26) | 0.1 (0.08) | 2.3 (0.90) | 1.8 (1.32) | 9.6 (1.71) | 12.3 (2.57) |
| Pak | 1 | 3.9 (0.43) | 8.3 (1.00) | 0 | 0.7 (0.40) | 1.1 (0.47) | 2.1 (0.31) | 0 | 0 | 1.3 (0.43) |
| | 2 | 4.0 (0.49) | 24.0 (9.13) | 0 | 12.4 (9.07) | 1.1 (0.47) | 4.0 (0.71) | 0.2 (0.11) | 0 | 5.2 (1.16) |
| | 3 | 4.8 (0.47) | 12.5 (1.28) | 0 | 0.9 (0.34) | 1.3 (0.49) | 3.1 (0.42) | 0.4 (0.23) | 0.2 (0.11) | 5.0 (0.79) |
| | 4 | 5.8 (0.33) | 22.1 (2.39) | 0.7 (0.28) | 0.4 (0.15) | 0.5 (0.15) | 2.8 (0.43) | 0.5 (0.23) | 0.8 (0.30) | 14.3 (1.87) |
| | 5 | 4.9 (0.58) | 20.4 (4.77) | 1.9 (0.82) | 0 | 0.1 (0.08) | 1.2 (0.51) | 0 | 7.2 (1.67) | 3.8 (0.95) |
| | 6 | 8.5 (0.62) | 44.5 (5.33) | 0.9 (0.38) | 0.1 (0.08) | 0.3 (0.13) | 5.8 (0.71) | 0.2 (0.11) | 6.9 (1.15) | 15.3 (4.07) |
| Par | 1 | 6.9 (0.58) | 240.7 (47.51) | 0.2 (0.11) | 58.1 (12.57) | 0.2 (0.11) | 17.8 (3.88) | 142.2 (35.30) | 0.1 (0.08) | 17.6 (3.87) |
| | 2 | 5.9 (0.50) | 125.7 (34.61) | 10.6 (7.23) | 55.1 (21.20) | 1.8 (0.89) | 4.0 (1.25) | 36.5 (18.33) | 1.8 (0.74) | 13.9 (2.84) |
| | 3 | 6.3 (0.39) | 85.6 (31.37) | 0.3 (0.19) | 40.7 (18.54) | 2.5 (0.76) | 2.7 (0.28) | 18.7 (9.22) | 1.5 (0.93) | 17.5 (6.10) |
| | 4 | 7.7 (0.57) | 71.8 (12.13) | 10.3 (1.85) | 0.2 (0.17) | 0.1 (0.08) | 4.0 (0.60) | 8.7 (7.15) | 36.2 (4.30) | 5.0 (2.00) |
| | 5 | 8.5 (0.63) | 46.0 (6.53) | 3.8 (0.84) | 0 | 0 | 3.7 (0.58) | 5.1 (2.05) | 19.9 (3.35) | 6.1 (1.58) |
| | 6 | 9.8 (0.72) | 55.6 (7.06) | 4.2 (1.19) | 0.9 (0.57) | 0.6 (0.23) | 2.8 (0.80) | 7.8 (2.71) | 13.6 (3.44) | 9.6 (2.13) |
| Wai | 1 | 3.0 (0.46) | 6.1 (0.73) | 0 | 0.6 (0.19) | 2.9 (0.53) | 0.2 (0.11) | 0.1 (0.08) | 0 | 0.1 (0.08) |
| | 2 | 4.6 (0.71) | 22.1 (5.10) | 3.6 (0.85) | 0.1 (0.08) | 0.7 (0.28) | 0.3 (0.19) | 1.0 (0.62) | 0 | 0.3 (0.19) |
| | 3 | 7.3 (0.48) | 24.7 (3.34) | 2.2 (1.21) | 0 | 0.3 (0.14) | 0.6 (0.19) | 0.5 (0.23) | 0.3 (0.13) | 0.3 (0.14) |
| | 4 | 9.5 (0.74) | 44.8 (2.85) | 8.5 (1.69) | 0.1 (0.08) | 0.6 (0.26) | 1.5 (0.47) | 2.8 (1.06) | 0.3 (0.14) | 0.5 (0.23) |
| | 5 | 5.5 (0.68) | 29.9 (7.85) | 4.8 (1.59) | 9.6 (7.30) | 1.7 (0.78) | 0.2 (0.17) | 4.4 (2.39) | 0.6 (0.34) | 1.3 (0.48) |
| | 6 | 12.8 (0.77) | 56.3 (6.46) | 5.2 (1.15) | 0 | 0.3 (0.13) | 1.5 (0.23) | 0.3 (0.33) | 1.6 (0.56) | 0.6 (0.23) |
| *1997* | | | | | | | | | | |
| Hel | 3 | 9.9 (0.50) | 261.8 (39.05) | 4.1 (2.32) | 141.0 (29.58) | 4.1 (1.00) | 5.1 (1.95) | 69.5 (21.02) | 1.7 (0.72) | 25.3 (4.73) |
| | 5 | 11.1 (0.71) | 66.1 (14.05) | 5.0 (1.31) | 6.7 (3.70) | 0.9 (0.29) | 4.2 (0.94) | 12.0 (4.64) | 4.3 (0.93) | 21.4 (5.18) |
| Pak | 2 | 7.8 (0.72) | 276.1 (35.77) | 0 | 140.3 (18.38) | 2.0 (0.46) | 2.3 (0.40) | 119.7 (41.47) | 0.2 (0.11) | 3.8 (1.21) |
| | 5 | 8.9 (0.53) | 88.6 (24.27) | 5.4 (1.03) | 0.3 (0.18) | 0.4 (0.19) | 2.5 (0.42) | 5.4 (2.24) | 20.0 (5.51) | 21.8 (4.91) |
| Par | 3 | 5.9 (0.47) | 155.6 (22.98) | 1.1 (0.56) | 129.9 (21.95) | 6.7 (1.12) | 6.3 (1.15) | 7.1 (4.88) | 0.1 (0.08) | 2.1 (0.68) |
| | 6 | 11.8 (0.97) | 139.2 (14.84) | 2.4 (1.01) | 0 | 0.1 (0.08) | 15.3 (3.35) | 14.8 (6.79) | 49.2 (9.21) | 16.8 (4.19) |
| Wai | 4 | 7.4 (0.54) | 33.1 (2.49) | 11.8 (1.04) | 0.3 (0.18) | 0.7 (0.33) | 0.9 (0.34) | 0.3 (0.18) | 0.2 (0.11) | 0.4 (0.19) |
| | 6 | 11.8 (0.43) | 48.3 (5.00) | 11.0 (1.99) | 1.2 (1.17) | 0.4 (0.34) | 0.3 (0.13) | 3.9 (1.45) | 0.5 (0.19) | 0.7 (0.38) |

ANOVA of numbers of individuals were not powerful enough to identify significant differences among estuaries (Table 4). Numbers of both taxa and individuals differed significantly within estuaries at spatial scales ranging from metres to hundreds of metres (i.e. among replicate cores within plots, plots within sites and sites within estuaries).

In 1995/1996, the total numbers of individuals (summed across all taxa) generally increased with increasing distance down the estuary in Hellyers, Pakuranga and Waiheke (Table 5). The converse pattern was evident in Paremoremo, where the greatest numbers of individuals were recorded at sites at the top of the estuary, with numbers progressively declining further down the estuary. The communities in Paremoremo

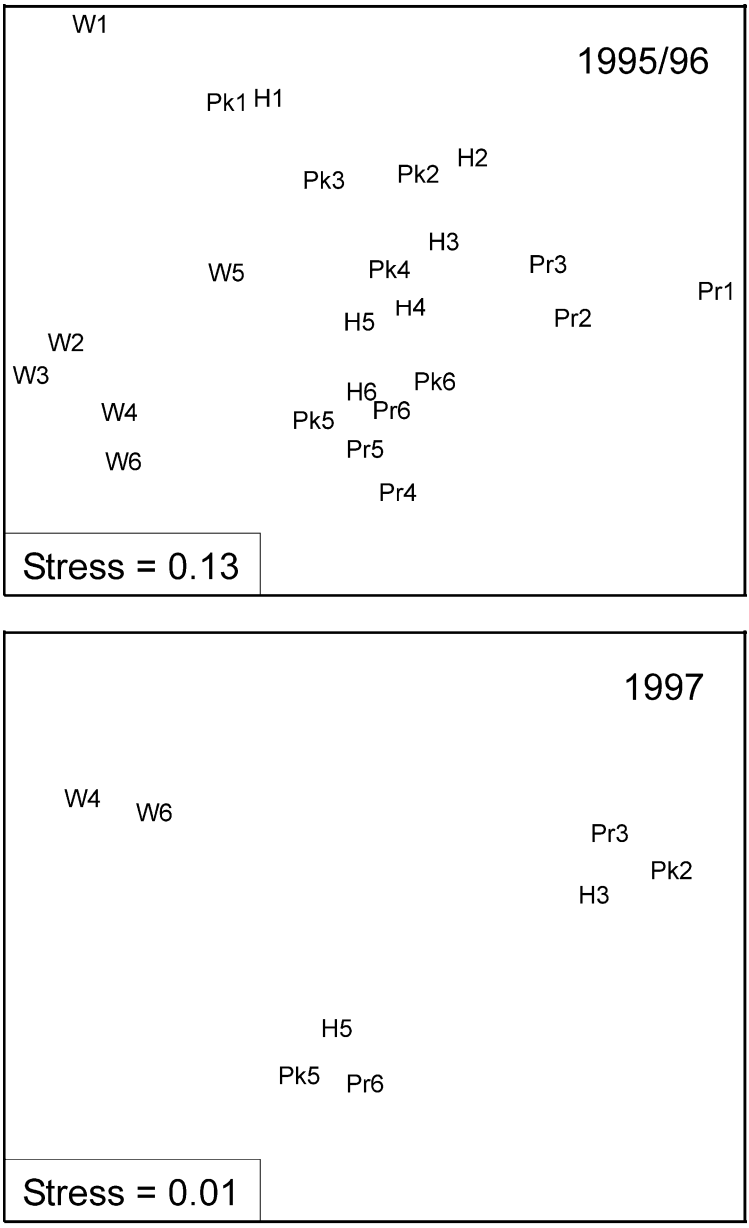


Fig. 2. NMDS plot of fauna from each of the sites in each of the four estuaries for the 1995/1996 (six sites sampled in each estuary, numbered 1–6) and 1997 (two sites sampled in each estuary) times of sampling. Individuals in each taxon were summed across the 12 core samples collected at each site for the purpose of these analyses. 'H' = Hellyers, 'Pk' = Pakuranga, 'Pr' = Paremoremo and 'W' = Waiheke.

were dominated numerically by oligochaetes and the amphipod corophiid sp. 1, both of which decreased in abundance downstream (Table 5). In 1997, the numbers of individuals were generally higher than those recorded in 1995/1996 (Table 5), and greater numbers of individuals were recorded at sites higher up the estuary than at sites further down in Hellyers, Pakuranga and Paremoremo. Numbers of taxa generally increased downstream in both 1995/1996 and 1997 and were larger in 1997 than in 1995/1996 (Table 5).

These results suggest that there were gradients in community composition along the estuaries. Furthermore, differences among times at the same site were at least as large as differences among sites, with corophiids being the taxa mainly responsible for this variation (Table 5). More taxa and individuals characterised communities at sites in the rural estuaries.

NMDS ordination plots indicated that faunal communities in samples from Waiheke were distinctly different from those from the other three sites in both 1995/1996 and 1997 (Fig. 2). In 1995/1996, samples from Paremoremo were also different from those of Hellyers and Pakuranga (Fig. 2). The 1995/1996 plot also showed a separation among sites along the estuarine gradients. ANOSIM showed that there were significant differences in community composition among the four estuaries in 1995/1996 ($R = 0.496$, $P = < 0.001$). The pairwise tests indicated that, in the 1995/1996 sampling period, the benthic communities from Waiheke were significantly different from those in the other three estuaries. Multiple comparison tests also indicated differences among Hellyers, Pakuranga and Paremoremo (Table 6) but, in the absence of formal correction for multiple comparisons, these should be interpreted cautiously (Clarke, 1999). The SIMPER analyses indicated that the same taxa were often important in contributing to differences among estuaries (Table 7). Since these taxa were generally present in all estuaries (Table 5), their role in identifying differences derives from differences in relative abundance among estuaries. Comparison of Tables 5 and 7 does not, however, indicate any obvious patterns of differences among estuaries, other than the fact that several taxa were relatively more abundant in Paremoremo.

Table 6
Results of 2-way ANOSIM tests of differences in benthic invertebrate communities among the four estuaries and among sites within estuaries in 1995/1996

| Comparison | 1995/96 | |
|---|---|---|
| | *R* | *P* |
| Among estuaries | 0.496 | < 0.0001 |
| Hel vs Pak | 0.026 | 0.335 |
| Hel vs Par | 0.270 | 0.037 |
| Hel vs Wai | 0.669 | 0.002 |
| Pak vs Par | 0.456 | 0.004 |
| Pak vs Wai | 0.800 | 0.002 |
| Par vs Wai | 0.719 | 0.002 |
| Among sites within estuaries | 0.532 | < 0.0001 |

Table 7
Summary of results of SIMPER analyses [19] showing average Bray–Curtis% dissimilarities among estuaries and the five taxa contributing most to these dissimilarities

| Estuaries compared | % Dissimilarity | | | | | |
|---|---|---|---|---|---|---|
| Hel vs Pak | 66.1 | Polyd | Cor 1 | Para 1 | Cos 1 | Ner |
| Hel vs Par | 74.3 | Oligo | Cor 1 | Para 1 | Polyd | Cap |
| Hel vs Wai | 87.0 | Prio | Polyd | Cor 1 | Cap | Hel/Para 1 |
| Pak vs Par | 76.6 | Oligo | Cor 1 | Para 1 | Polyd | Cap |
| Pak vs Wai | 88.4 | Prio | Polyd | Cap | Ner | Cor 1 |
| Par vs Wai | 88.9 | Oligo | Cor 1 | Para 1 | Prio | Polyd |

Dissimilarities are averaged across all comparisons between plots from the two estuaries being compared (cores pooled). Names of estuaries are abbreviated as in Table 2. Taxon abbreviations: Arth, *Arthritica bifurca;* Cos, *Cossura* sp. 1; Para, Paraonid sp. 1; Aust, *Austrovenus stutchburyi;* Hel, *Helice crassa;* Polyd, polydorid spionids; Cap, capitellids; Ner, nereids; Prio; *Prionospio aucklandica;* Cor 1, Corophiid amphipod sp. 1; Oligo, oligochaetes; Theo, *Theora lubrica*

NMDS ordination plots (Fig. 3, for 1995/1996) and ANOSIM tests (Table 8 for 1995/1996 and 1997) for differences among sites within each estuary (plot = replicate; cores pooled) showed that there were differences in community composition among sites in each of the estuaries in 1995/1996 and 1997. The differences were most marked at Pakuranga and Paremoremo as indicated by the higher values of the $R$ statistic (Table 8). The ordination plots for 1995/1996 suggest that, rather than distinct communities characterising each site within each estuary, there are gradients in macrofaunal community composition, with sites that are close together being more similar in community composition than sites that are further apart. As with comparisons among estuaries, a small suite of taxa contributed most to differences among sites within estuaries (Table 9).

For all four estuaries, the two-way crossed ANOSIM tests comparing among sites within estuaries and among times of sampling, indicated a significant effect of both sampling time and site within the estuary (Table 10). Thus, in addition to the differences in benthic community composition among sites within an estuary, there were also significant differences in benthic community composition at the different sampling times.

### 3.3. Relationships between distributions of physical and chemical variables and those of benthic invertebrates

The distributions of benthic invertebrate were significantly related to those of environmental variables (Monte Carlo permutation test: $F = 4.08$, $P = 0.01$ in 1995/1996 and $F = 4.51$, $P = 0.01$ in 1997). The importance of the first and second ordination axes in describing the relationship between benthic communities and environmental variables was shown by the taxa-environment correlation values (0.907 and 0.931 for the first and second axes, respectively, in 1995/1996 and 0.966 and 0.889 in 1997). Their importance was also indicated by the eigenvalues (a measure of how well the taxa and the samples in which they occur correspond in position along

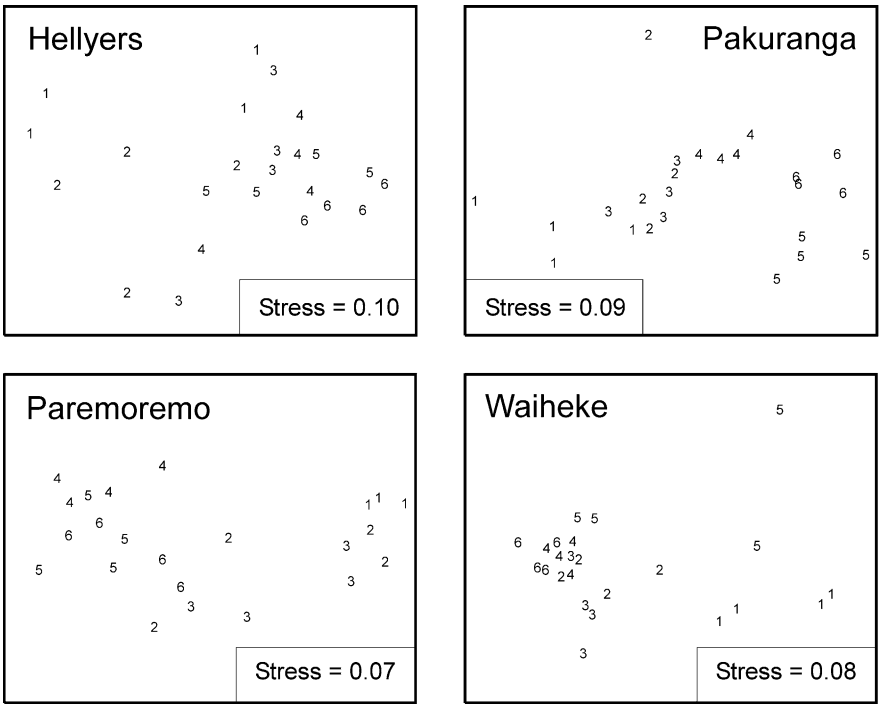


Fig. 3. NMDS plots of fauna from each of the six sites in each of the four estuaries for the 1995/1996 time of sampling. Replicates are plots and individuals in each taxon were summed across the three core samples collected at each plot for the purpose of these analyses. Sites are labelled 1–6.

Table 8
Results of separate 1-way ANOSIM tests for differences among sites within each of the four estuaries at the 1995/1996 and 1997 times of sampling

| Estuary | 1996/1995 | | 1997 | |
|---|---|---|---|---|
| | *R* | *p* | *R* | *p* |
| Hellyers | 0.305 | <0.001 | 0.938 | 0.029 |
| Pakuranga | 0.704 | <0.001 | 1.000 | 0.029 |
| Paremoremo | 0.618 | <0.001 | 1.000 | 0.029 |
| Waiheke | 0.500 | <0.001 | 0.719 | 0.029 |

Faunal abundance data were untransformed and data for each taxon were summed across cores for each plot.

the axes) of axis 1 (0.482 in 1995/1996, 0.592 in 1997) and axis 2 (0.264 in 1995/1996, 0.264 in 1997). These contrast with the relatively small values for the third axis (0.172 in 1995/1996, 0.103 in 1997). In the 1995/1996 data set, the first axis (which accounted for 37.9% of the variance of the taxa-environment relationship) was positively correlated with % sand (Table 11). The sediment concentrations of the

Table 9
Summary of results of SIMPER analyses showing average Bray–Curtis % dissimilarities among sites within estuaries, based on their faunas

| Estuary | Sites compared | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 1 vs 2 | 1 vs 3 | 1 vs 4 | 1 vs 5 | 1 vs 6 | 2 vs 3 | 2 vs 4 | 2 vs 5 | 2 vs 6 | 3 vs4 | 3 vs 5 | 3 vs 6 | 4 vs 5 | 4 vs 6 | 5 vs 6 |
| Hel | 70.4 | 76.0 | 79.0 | 77.8 | 87.0 | 71.7 | 73.0 | 70.4 | 93.0 | 59.1 | 60.6 | 68.5 | 54.1 | 59.3 | 49.7 |
| | Cor 1 | Cor 1 | Polyd | Polyd | Polyd | Cor 1 | Cor 1 | Cor 1 | Cor 1 | Cor 1 | Cor 1 | Cor 1 | Polyd | Polyd | Polyd |
| | Hel | Polyd | Cor 1 | Cos 1 | Para 1 | Polyd | Polyd | Polyd | Para 1 | Polyd | Polyd | Para 1 | Cor 1 | Cap | Cos 1 |
| | Polyd | Hel | Ner | Para 1 | Cos 1 | Hel | Hel | Hel | Polyd | Ner | Cos 1 | Polyd | Cos 1 | Para 1 | Para 1 |
| Pak | 61.2 | 54.8 | 70.9 | 80.7 | 84.1 | 42.4 | 55.5 | 74.7 | 68.4 | 48.1 | 68.8 | 67.0 | 68.0 | 47.0 | 48.9 |
| | Cor 1 | Polyd | Polyd | Para 1 | Polyd | Cor 1 | Polyd | Para 1 | Polyd | Polyd | Para 1 | Polyd | Polyd | Cos 1 | Polyd |
| | Polyd | Arth | Arth | Cos 1 | Cos 1 | Polyd | Cor 1 | Cor 1 | Cos 1 | Hel | Cos 1 | Cos 1 | Para 1 | Para 1 | Cos 1 |
| | Ner | Ner | Hel | Polyd | Para 1 | Ner | Ner | Cos 1 | Cor 1 | Ner | Ner | Para 1 | Cos 1 | Polyd | Ner |
| Par | 57.4 | 61.0 | 88.5 | 89.1 | 85.0 | 52.6 | 75.8 | 73.4 | 68.7 | 80.2 | 72.8 | 66.0 | 43.3 | 51.1 | 43.9 |
| | Oligo | Oligo | Oligo | Oligo | Oligo | Cor 1 | Cor 1 | Cor 1 | Cor 1 | Para 1 | Cor 1 | Cor 1 | Para | Para 1 | Para 1 |
| | Cor 1 | Cor 1 | Cor 1 | Cor 1 | Cor 1 | Oligo | Para 1 | Oligo | Oligo | Cor 1 | Para 1 | Para 1 | Oligo | Oligo | Cor 2 |
| | Ner | Ner | Para 1 | Para 1 | Ner | Polyd | Oligo | Para 1 | Para 1 | Oligo | Oligo | Oligo | Cap | Cor 2 | Oligo |
| Wai | 81.4 | 87.0 | 90.3 | 83.2 | 93.5 | 52.1 | 46.5 | 71.0 | 55.5 | 49.9 | 74.9 | 55.0 | 69.0 | 38.9 | 76.5 |
| | Prio | Prio | Prio | Cor 1 | Prio | Prio | Prio | Prio | Theo | Prio | Prio | Theo | Prio | Theo | Prio |
| | Cap | Aust | Cap | Cap | Theo | Aust | Cap | Cor 1 | Prio | Cap | Cor 1 | Prio | Cor 1 | Prio | Theo |
| | Hel | Hel | Theo | Prio | Cap | Cap | Theo | Cap | Cap | Aust | Cap | Cap | Cap | Cap | Cor 1 |

The three taxa contributing most to these dissimilarities are shown. Dissimilarities are averaged across comparisons of all samples from each site. Abbreviations of names of estuaries are as in Table 2 and taxa as in Table 7.

organic contaminants dieldrin, chlordane and PCB, and % gravel, were also all positively correlated with axis 1. Percent VOM, % distance along the estuary, % dry weight, and concentrations of Fe and Cu were negatively correlated with axis 1 (Table 11). The second axis (which accounted for 20.8% of the variance of the taxa-environment relation) was positively related with the concentrations of organic (chlordane, PCB, PAH, dieldrin) and inorganic (Cu, Pb, Zn) contaminants, sediment consistency and % gravel. It was negatively correlated with % distance along the estuaries (Table 11). In the 1997 data set, % distance along the estuary had the strongest (negative) correlation with the first axis (Table 11), which accounted for 51.4% of the taxa-environment variation. Pb concentration and % VOM were also negatively correlated with axis 1, while sediment % dry weight was positively corre-

Table 10
Results of the two-way ANOSIM tests for effects of sampling time and site using untransformed benthic invertebrate abundance data from each of the four estuaries

| Estuary | Sampling time | | Site | |
|---|---|---|---|---|
| | *R* | *P* | *R* | *P* |
| Hellyers | 0.479 | 0.002 | 0.495 | 0.011 |
| Pakuranga | 0.833 | 0.001 | 0.891 | 0.001 |
| Paremoremo | 0.391 | 0.015 | 0.823 | 0.001 |
| Waiheke | 0.833 | 0.001 | 0.596 | 0.001 |

Table 11
Interset correlations of the environmental variables with the first two species axes of the CANOCO plots for the 1995/1996 and 1997 data

| Variable | 1995/1996 | | 1997 | |
|---|---|---|---|---|
| | axis 1 | axis 2 | axis 1 | axis 2 |
| % gravel | **0.2149** | **0.2475** | −0.1012 | −0.0549 |
| % sand | **0.7942** | −0.0631 | [a] | [a] |
| % mud | [a] | [a] | [a] | [a] |
| Dry weight | **−0.5183** | 0.1264 | **0.5532** | −0.0531 |
| % VOM | **−0.7473** | −0.0555 | **−0.4530** | 0.2988 |
| consistency | −0.1045 | **0.3137** | −0.1415 | −0.2378 |
| sediment Cu | **−0.2541** | **0.5729** | [a] | [a] |
| sediment Zn | 0.1594 | **0.6537** | −0.2334 | −0.2707 |
| sediment Pb | 0.0410 | **0.6678** | **−0.5037** | 0.1561 |
| sediment Fe | **−0.2859** | −0.1105 | 0.1218 | **0.4728** |
| PAH | 0.0931 | **0.6701** | [b] | [b] |
| Dieldrin | **0.3756** | **0.5495** | [b] | [b] |
| Chlordane | **0.2673** | **0.7479** | [b] | [b] |
| DDT | 0.0272 | 0.0328 | [b] | [b] |
| PCB | **0.2620** | **0.7313** | [b] | [b] |
| % distance | **−0.7449** | **−0.2776** | **−0.7349** | 0.3122 |

For descriptions of the variables, see text. Correlations significant at the 5% level are shown in bold.

[a] Environmental variable omitted due to high inflation factor. A high inflation factor occurs when an environmental variable is almost perfectly correlated with other variables and thus has no unique contribution to the regression equation. These variables are unstable and therefore have no interpretative value.

[b] Environmental variable not measured in 1997.

lated. Fe concentration had the highest correlation with axis 2, which explained 21.8% of the variation in the taxa–environment relation.

Sample-environment ordinations based on partial CCA of all the sites in all four estuaries for both the 1995/1996 and 1997 data (Fig. 4a and b) indicated that there was a clear and consistent separation of the benthic communities at different sites along each of the estuaries. Sites high up the estuaries (1–3) were separated from sites lower down the estuaries (4–6) along axis 1, which was strongly correlated with sediment variables (e.g. % gravel, % sand, % VOM), % distance and organic contaminants, and also along axis 2 which was generally correlated with sediment contaminants (organic and metals). Such correlations are not necessarily proof of a cause–effect relationship between these particular contaminants/environmental variables and faunal distributions, but they are indicative of the presence of gradients of environmental variables within estuaries that may be influencing the composition of benthic communities. For example, in 1995/1996, sites were separated along axis 2 (strongly associated with concentrations of contaminants in sediments) into those sites in the upper parts of the estuaries which have higher levels of contamination than those sites further down the estuary. There was generally no clear separation of the estuaries in either of the ordination biplots, except for the case of Pakuranga and Paremoremo.

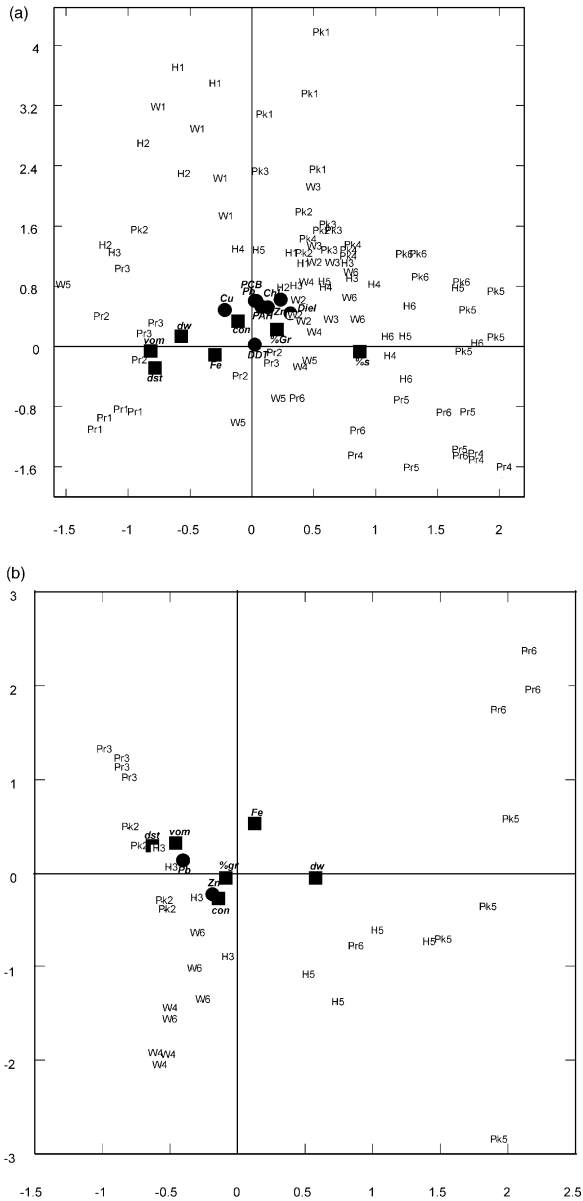


Fig. 4. Sample-environment plots from CANOCO for data from (a) 1995/1996 and (b) 1997. Squares represent environmental variables ('%s' = % sand in sediment; '%gr' = % gravel in sediment; 'vom' = % volatile organic matter content of sediment; 'dst' = distance of site from mouth of estuary as% of total length of estuary; 'dw' = dry weight of sediment as% of wet weight; 'con' = consistency of sediment) and circles represent contaminant concentrations ('Cu', 'Pb', 'Zn', 'Fe', 'Chl', 'DDT', 'Diel', 'PAH' and 'PCB' are concentrations of Cu, Pb, Zn, Fe, chlordane, DDT, dieldrin, PAH and PCB, respectively). Names of estuaries are abbreviated as in Fig. 2 and site numbers (1–6) are written beside each abbreviation. See text for details of variables and analyses.

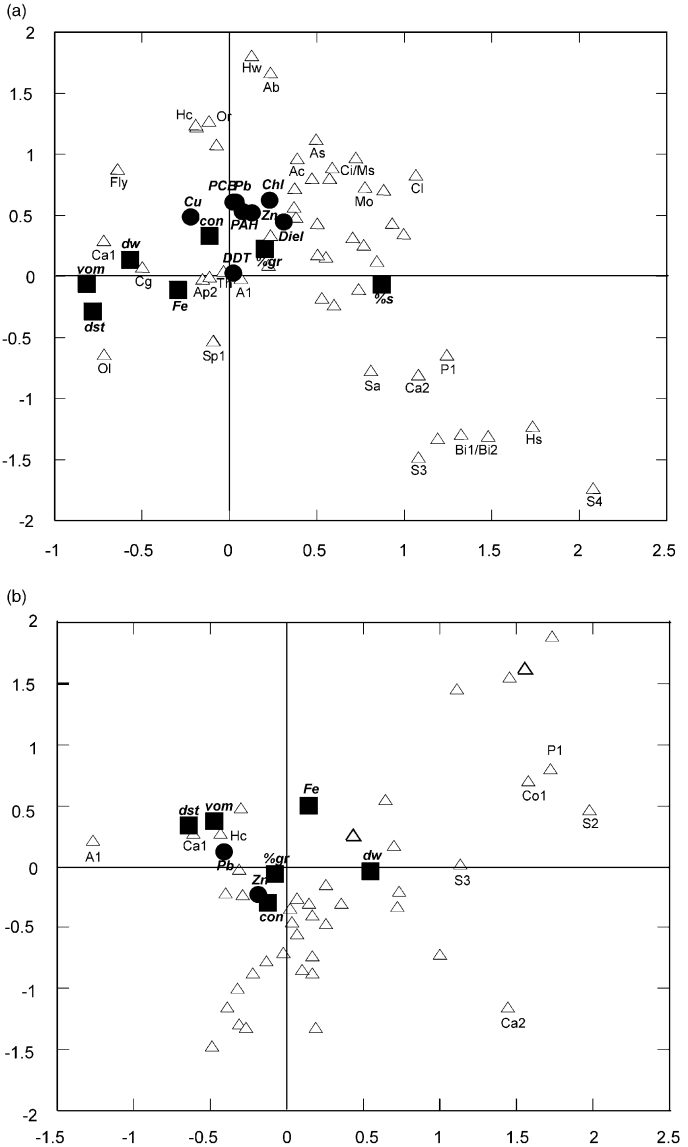


Fig. 5. Taxa–environment plots from CANOCO for data from (a) 1995/1996 and (b) 1997. Squares represent environmental variables, circles represent contaminants (see Fig. 4 for explanation of labels) and triangles represent taxa. Representative taxa, including those discussed in the text, are labelled for illustrative purposes. Abbreviations of names of taxa are: A1, *Aonides* sp. 1; Ab, *Arthritica bifurca*; Ac, *Amphibola crenata*; Ap2, amphipod sp. 2; As, *Austrovenus stutchburyi*; Bi1, bivalve sp. 1; Bi2, bivalve sp. 2; Ca1, corophiid amphipod sp. 1; Ca2, corophiid amphipod sp. 2; Cg, *Cominella glandiformis*; Ci, cirrolanid isopod; Cl, *Colurostylis lemurum*; Co1, *Cossura* sp. 1; Fly, dipteran larva; Hc, *Helice crassa*; Hs, *Hiatula siliqua*; Hw, *Halicarcinus whitei*; Mo, *Mactra ovata*; Ms, *Macroclymenella stewartensis*; Ol, oligochaetes; Or, orbinid sp. 1; P1, paraonid sp. 1; Sa, sabellids; S3, syllid sp. 3; S4, syllid sp. 4; Sp1, spionid sp. 1; Th *Torridoharpinia hurleyi*. See text for details of variables and analyses.

Examination of the taxa-environment biplot for 1995/1996 (Fig. 5a) indicates that some taxa were found predominantly at sites high up the estuaries, in sediments with a high % VOM content, a large % dry weight and a low % sand; whereas others were more characteristic of benthic communities further down the estuaries, where the sediments were characterised by high % sand and low % VOM content. The distributions of some taxa could also be related to concentrations of sediment contaminants (or some unmeasured environmental variable which was highly correlated with contaminant levels). The crab *Halicarcinus whitei*, the bivalves *Arthritica bifurca* and *Austrovenus stutchburyi* and the gastropod *Amphibola crenata*, for example, were most abundant in sediments with high concentrations of organic and metal contaminants. Oligochaetes and spionid polychaete sp. 1 were commonly found in sediment where contaminant concentrations were low. In 1997, similar patterns of distribution of taxa were evident (Fig. 5b).

## 4. Discussion

Concentrations of contaminants were larger in sediments from the two urbanised estuaries than the non-urbanised ones. There were, however, a few unexpected results, particularly the relatively high concentrations of DDT in sediments in non-urbanised Paremoremo. This may be a consequence of past use of this chemical as a pesticide in parts of the catchment used for horticulture. Between the two non-urbanised estuaries, concentrations of contaminants were generally higher in sediments from Paremoremo than those from Waiheke. This is consistent with the relative distance of the two estuaries from external sources of contamination, namely nearby estuaries with urbanised catchments (Fig. 1). Although none of the contaminants exceeded the effects range-median (ER-M) guideline concentrations (Long et al., 1995), the effects range-low (ER-L) concentrations were exceeded by Pb at some sites in Hellyers and Pakuranga Creeks, Zn in Pakuranga and DDT in Hellyers, Pakuranga and Paremoremo. The ER-L values represent concentrations at which biological effects may possibly occur and indicate a potential for effects in the estuaries studied here. Effects range concentrations refer to $HNO_3$-extracted metals, whereas the extractant in the present study was HCl. Empirically-derived correction factors (10, 10 and 37 $\mu g\ g^{-1}$ for Cu, Pb and Zn, respectively: Williamson, unpublished data) were applied to the concentrations found in the present study. Again, none of the concentrations exceeded ER-M values, but concentrations of Cu at Site 1 in Hellyers and Sites 1–3 in Pakuranga, Pb at Sites 1–3 in both Hellyers and Pakuranga, and Zn at Sites 1–6 in Pakuranga exceeded the ER-L values. ER-L values were not exceeded at any sites in Paremoremo or Waiheke. The fact that concentrations of metals in porewaters were below the limits of detection suggests that this is not likely to be a major route of exposure for the organisms living in the sediments.

Gradients in environmental variables along the estuaries were paralleled by variation in the abundance and diversity of invertebrates. Patterns of distribution varied among taxa, with different taxa achieving maximal abundances at different

points along the estuaries. Comparisons of macrofaunal communities among the estuaries using NMDS and ANOSIM (comparing relative values of the $R$ statistic for the different comparisons; Clarke, 1999) showed that those in Hellyers and Pakuranga were relatively similar to each other and differed from those in the non-urbanised estuaries, particularly Waiheke (Table 6). The communities in the two non-urbanised estuaries were also different from each other. This similarity between the faunas of the two urbanised estuaries and their dissimilarity from the faunas of the non-urbanised estuaries is consistent with an impact of contaminants derived from urban runoff. If differences among the faunas of the different estuaries were determined solely by natural environmental factors, one might have expected greatest similarity between the faunas of Paremoremo and Hellyers, given their geographical proximity (Fig. 1), but this was not the case.

Further evidence consistent with the role of contaminants in structuring benthic communities was provided by the CANOCO analyses. Samples and taxa were arranged along axes correlating with gradients of natural environmental variables and contaminants, these two groups of variables tending to correlate with different axes. In these analyses, there were no clear differences among estuaries with urbanised and non-urbanised catchments, but this might be expected if the main environmental factors influencing benthic communities were natural ones, given that the sites were chosen so that their positions along these gradients were likely to overlap among the estuaries. The effects of contaminants would then be expected to emerge as modifications of patterns derived from these natural environmental variables. Watzin et al. (1997) reported that physical environmental variables were as important as contaminant concentrations in predicting the structure of benthic communities in a moderately contaminated lake (concentrations of metals in lake sediments were generally $>$ lowest effects levels for freshwater sediments (Persaud, Jaagumagi, & Hayton, 1993) but $<$severe effects levels while, in many cases, concentrations of organic contaminants were $>$ severe effects levels). Similarly, sediment grain-size was the best predictor of amphipod mortality in tests with moderately contaminated sediments from adjacent to a coastal wastewater outfall in California (Maxon et al., 1997). Addition of contaminant concentrations to the stepwise regression model did not improve predictions.

Detection of effects of contaminants in the highly variable estuarine environment is difficult even when concentrations are much higher than in the present study. Diaz (1989) attempted to reduce the confounding effect of the salinity gradient on correlations between biota and contaminants in the James River Estuary, Virginia, by analysing data from different sections of the gradient separately. In the saline portions of the estuary, abundances and biomass of dominant taxa were larger immediately below a large sewage outfall but otherwise the community was similar to that found away from the outfall (Diaz, 1989). In contrast, in the tidal, freshwater part of the estuary, where the confounding influences of salinity and estuarine circulation were absent, the fauna showed more widespread changes that correlated with contamination. Communities showed least diversity in areas where outfalls were most abundant. Contrary to the above study, Schlekat et al. (1994) found no significant correlations among abundances of taxa or individuals and various sediment-related

variables, including concentrations of organic and heavy-metal contaminants, in their study of tidal rivers around Washington, DC. Concentrations of contaminants in these rivers ranged from 13.7 to 348, 32 to 3630 and 62.0 to 1090 μg $g^{-1}$ for Cu, Pb and Zn, respectively (reported in Velinsky et al., 1994) and derived from storm and combined sewers and (during overflow) industrial sources. These concentrations include values considerably larger than those found in our study.

Numbers of taxa and individuals were generally larger in 1997 than in 1995/1996 for those sites (two per estuary) that were sampled both times (Table 5). In 1995/1996, sites tended to separate along the ordination axes in patterns that correlated with sediment texture and concentrations of several organic contaminants (Fig. 4a, axis 1) or organic and inorganic contaminants (axis 2). In 1997, in contrast, the correlations with sediment texture and concentration of Zn were not observed, but the correlation with concentration of Pb was still present (Fig. 4b: note that concentrations of organic contaminants were not measured in 1997 and that Cu was removed from the analysis for 1997 because it was highly correlated with other environmental factors). Comparing the correlations of individual taxa with concentrations of heavy metals, as identified by the ordination axes, between 1995/1996 and 1997 (Fig. 5) showed consistent relationships for some of the taxa (e.g. *Helice crassa* and the polychaete orbiniid sp. 1), correlation one time but not the other, or a change in the correlation from negative to positive (e.g. oligochaetes and sabellid polychaetes). There was reasonable consistency in the environmental variables that correlated with ordination axes between the two sampling times (Table 11), notable exceptions being the lack of correlation of sediment texture and concentration of Zn in 1997.

These results indicate that relationships between macrofaunal communities and suites of environmental variables in estuaries may change through time. Temporal variation in the suite of predictor environmental variables for benthic community structure was reported by Watzin et al. (1997) in their study of a moderately contaminated lake. There is currently interest among estuarine ecologists in using ordination techniques to develop empirical relationships between estuarine faunas and sets of natural environmental variables as a tool to identify effects of contaminants, equivalent to methods such as the River Invertebrate Prediction and Classifications System (Wright, 1995) and the Benthic Assessment of Sediment (Reynoldson, Bailey, Day, & Norris, 1995) for freshwaters. Given the spatially and temporally variable nature of estuarine environments, this interest has been tempered with caution, and the temporal variation in the relationships identified in the present study suggests that this is wise.

While the present study has provided some correlational evidence consistent with a biological effect of contamination derived from stormwater runoff, it suffers from the problem of demonstrating cause and effect that is common to all such studies. Measurement of other, natural stressors, such as concentrations of dissolved oxygen, ammonia and sulphides and redox potential, might have helped clarify the role of contaminants from stormwater. Integrative assessments of impact based on weight-of-evidence drawn from patterns of distribution of contaminants and organisms in the field and toxicity tests (Chapman, Power, & Burton, 1992) provide

perhaps the most realistic method of identifying impacts of contaminants derived from stormwater on estuarine macrofauna (Watzin et al., 1997). The correlative approach is thereby made more robust by the additional use of toxicity tests (or other forms of manipulative experiments). The former may provide evidence, in the form of patterns of distribution of contaminants and fauna, *consistent* with an impact. In the absence of information on the distribution of contaminants, on the other hand, experimental approaches cannot demonstrate an impact in a particular estuary. Their value lies in testing the causal relationship underlying the correlations.

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